

“MAGA” The Ultimate Raspberry Pi RetroArch Build

Nearly 180GB of Retro Gaming, Computers, Arcade, EmulationStation & More

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This is not another Raspberry Pi image containing a frontend, a pile of folders and a RetroArch installation.

This is the sort of Raspberry Pi retro gaming project that reminds us why we got into this hobby in the first place, built around Debian 13 Trixie, a current RetroArch 1.22.2-based configuration and a seriously impressive collection of carefully organized retro gaming and computing content, this Raspberry Pi 4 and Raspberry Pi 5 build goes far beyond simply giving you a menu from which to launch games.

It is part retro console, part arcade machine, part vintage computer museum, part multimedia system and, for anyone who likes getting under the bonnet, part fully fledged Linux computer.

The package has evolved over several years and includes almost 128GB dedicated to emulated systems and related content, with approximately another 50GB of multimedia. It has been designed around Raspberry Pi hardware rather than simply copied from a generic Linux installation, with automatic model detection applying appropriate changes depending upon whether it is running on a Raspberry Pi 4/400 or Pi 5/500-class machine.

There are custom playlists. There is a separate modern EmulationStation experience. There are game manuals. There are controller profiles. There is support for RetroAchievements. There is NetPlay. There are attract videos. There are multiple arcade generations, consoles, handhelds and computers spanning decades.

And underneath all of it is a proper Linux environment for people who want considerably more than an appliance-style retro gaming box.

If you have a Raspberry Pi 4 or Raspberry Pi 5 looking for a purpose, this could be one of the most interesting images you put on it.

What Exactly Is This Raspberry Pi RetroArch Build?

The project is a customized plug-and-play RetroArch configuration built on a Debian 13 Trixie boot image. It is intended to operate on:

- Raspberry Pi 4 / 400
- Raspberry Pi 5 / 500
- Raspberry Pi 500+

The system can boot into either a Linux console-based gaming environment or a more traditional Wayland desktop.

That distinction is important.

A lot of Raspberry Pi retro images effectively turn the Pi into a single-purpose console. This one doesn't have to. It retains the flexibility of the underlying Linux system while providing a heavily customized front end that can be operated like a dedicated retro gaming machine.

The project has been continually developed since 2021, with the current image incorporating years of configuration changes, emulator updates, controller tweaks, playlists, artwork, manuals and usability improvements.

Why This Build Stands Out

Anyone can install RetroArch. That isn't the difficult bit.

The difficult bit is turning RetroArch into something that somebody can actually sit down in front of and enjoy without repeatedly disappearing into settings menus.

That means selecting appropriate emulator cores, mapping controllers, handling computer keyboards, changing disks, configuring overlays, choosing sensible display settings, dealing with arcade quirks, handling Bluetooth devices, providing usable menus and making dozens of completely different machines feel like parts of a single system.

That is where this build gets interesting. Highlights include:

- Debian 13 Trixie underneath the gaming environment
- RetroArch 1.22.2-based components with rebuilt emulator cores
- Automatic detection and configuration differences for Raspberry Pi 4/400 and Pi 5/500 hardware
- A custom console-based Playlists menu
- A separate EmulationStation kiosk-style frontend
- Support for arcade machines, home consoles, handhelds and vintage computers
- Approximately 128GB of systems/emulation content
- Approximately 50GB of additional multimedia

- Integrated game manual access
- RetroAchievements support
- RetroArch NetPlay support
- Save-state hotkeys
- Attract-mode videos
- Controller support ranging from simple retro pads to modern PlayStation controllers
- Enhanced PlayStation 5 DualSense integration
- Pi 500+ RGB keyboard integration and key highlighting where supported
- USB and Bluetooth controller detection
- Automatic game/computer keyboard focus handling
- Special mappings for systems such as Atari, ColecoVision, Commodore and DOS
- Game manuals accessible through the Bookshelf system
- Dolphin optimisations intended to reduce GameCube latency
- Current Stella development builds for Atari 2600 emulation
- Hypseus/Singe support for laserdisc titles
- Desktop access for anyone wanting to use the Raspberry Pi as a Linux computer as well

Two Ways To Explore The Collection

One of the nicest parts of the project is that users aren't locked into one frontend. There are essentially two major ways to get into the retro collection.

1. The Console Playlists Interface

This is the project's distinctive text-driven interface and is ideal when the Raspberry Pi is being treated as a dedicated retro machine.

It is designed for fast navigation and can be controlled using:

- Gamepad
- Keyboard
- Mouse

The menu provides access not only to systems but to carefully organized collections and quick-picks.

It also provides system functions without forcing users back into Linux.

From the console menu you can access things such as:

- Game playlists
- Quick-pick games
- Wireless status
- Connected USB controllers
- Connected Bluetooth controllers
- Master volume

- RetroAchievements credentials
- Bookshelf/manual searches
- System upgrade functionality
- Power off
- Reboot
- Desktop mode

There is even an idle attract mode which can cycle through videos and favorite games when the machine isn't being actively used.

For an arcade cabinet, games room or dedicated living-room Pi installation, that small detail makes a surprisingly big difference.

2. EmulationStation

If you prefer artwork, system carousels, game information and a more conventional console-style interface, the image also includes a customized EmulationStation setup.

EmulationStation operates as a simplified kiosk-style frontend and can be launched directly from the gaming environment.

Multiple updated themes are included:

- Animatics-DX
- PlayStation-X
- Retro_LUX
- iconic-es
- Hypermax-Plus-PixN

Game entries can include additional information and, importantly, manual links. Where the Manual icon is present, documentation can be fetched and viewed through the Bookshelf facility.

There are also familiar organizational views including:

- All Games
- Favorites
- Last Played
- Educational
- Pinball
- Racing
- Sports

A Proper Tour Through Retro Gaming History

The scope of the system is substantial.

This isn't a build centered exclusively around the 16-bit generation or a few arcade ROM sets. The active emulator configuration stretches from early Atari and Commodore hardware right through to considerably newer systems.

Arcade support uses a mixture of modern MAME, MAME 2003-Plus and FinalBurn Neo Light configurations where appropriate.

Among the curated arcade selections documented with the image are classics. There has also been specific work around analogue arcade games, MAME artwork, bezels and crosshair cursors for suitable games such as Crossbow and Sea Wolf. The current image moved back to a modern Stella 8.0pre standalone/core implementation rather than the older Stella 2014 configuration. Vintage Apple computer support is provided through MAME-based configurations. VIC-20 support uses VICE and includes a pleasing mixture of games, educational software and unusual early home-computer material.

The C64 is understandably well represented. Amiga 500, Amiga 1200 and CD32 support is handled through PUAE. This is also one of the areas where the build's specialized computer controls become especially useful, because the project provides direct handling for floppy-disk changes and keyboard focus rather than expecting everything to behave like a console.

The wonderfully distinctive vector-based Vectrex is supported through VecX. Laserdisc gaming gets its own place in the build through Hypseus/Singe support. Recent project updates specifically mention corrected Hypseus support, a default bezel and refreshed laserdisc handling. The NES/Famicom side uses Mesen and contains an excellent cross-section of the machine's library. PC Engine and CD support is provided through the Mednafen PCE core. SNES support can use Snes9x or bsnes-jg depending upon the performance/accuracy requirements and Raspberry Pi model.

And That Still Isn't Every Emulated Machine.

Quick-Picks Make A Huge Collection Less Intimidating

One problem with enormous retro collections is choice paralysis.

Thousands of entries sound brilliant until somebody sits down with a controller and spends twenty minutes scrolling.

This project addresses that with quick-picks and curated collections.

Rather than only organizing content by machine, the menus have evolved to include categories such as:

- Simple arcade favorites
- Board & Party
- Pinball
- RTS & Turn-Based
- Niche Controllers
- Homebrews
- Educational
- Racing
- Sports
- Favorites
- Last Played

That makes this far easier to hand to somebody who doesn't care which revision of MAME is being used and simply wants to play something good.

RetroAchievements Support

RetroAchievements – referred to as Cheevos within the interface – are supported for compatible games.

From the console interface, users can enter their achievement account credentials and use supported games to record accomplishments.

This is a brilliant addition for players who already know the classics inside out.

An old game suddenly gets a new layer of objectives, completion goals and competition without changing the original game itself.

The Bookshelf: Manuals Without Searching The Web

Another feature we particularly like is the Bookshelf.

One of the biggest headaches with old computer games is that a controller alone isn't always enough.

Some games were sold with:

- large instruction manuals
- keyboard reference cards
- maps
- command lists
- copy-protection material
- multi-page tutorials

The system provides manual searching and documentation integration, particularly for computer and console software.

The logo key can trigger a Bookshelf PDF manual search from the console interface, while supported EmulationStation entries can expose a Manual option and retrieve documentation through the Bookshelf system.

For genuine vintage-computing use, that is a far more important feature than it initially sounds.

Controller Support: RetroPad Makes Everything More Consistent

RetroArch uses its virtual RetroPad abstraction between your physical controller and the individual emulator core.

That means your real controller might be:

- a keyboard
- a simple USB arcade joystick
- a two-button retro controller
- an 8BitDo pad
- a PlayStation controller
- a modern DualSense controller

But RetroArch translates that physical input into a common virtual controller before the core maps it to the original machine.

This greatly simplifies switching from one system to another.

PlayStation 5 DualSense Support

The project has received specific work around Sony PlayStation 5 DualSense controller detection and integration, including earlier LED support.

For anybody planning a polished couch-gaming setup, using a current wireless controller with machines stretching back to the 1970s is a nice combination of old and new.

Pi 500+ RGB Keyboard Integration

One of the more unusual additions in the recent Debian 13 generation is support for Pi 500+ keyboard lighting.

Where possible, the configuration can highlight relevant keycaps used by the running content.

That is exactly the sort of small enthusiast feature that shows how far beyond a stock RetroArch installation this project has traveled.

Essential Console Menu Controls

The console Playlists menu is designed to work from a gamepad, keyboard or mouse.

Useful commands include:

- Left/Right – decrease or increase master volume where applicable.
- Controller back/Escape – back out of an option.
- Wireless option – view network status.
- Controller-list option – show connected USB and Bluetooth controllers.
- Quick-pick option – list simpler/popular game selections.
- Left Control – access RetroAchievements credential assignment.
- Logo key – trigger Bookshelf manual searching.
- Useful RetroArch Keyboard Hotkeys

The supplied configuration also defines keyboard shortcuts for common operations.

Important examples include:

- Caps Lock – toggle Game Focus / keyboard mode.
- Print Screen – take a screenshot.
- Right Shift / Select – Select or Option depending upon system.
- Enter / Start – Player 1 Start.
- K – save/keep a save state.
- L – load a save state.
- N – next save-state slot.
- P – previous save-state slot.

- , and . – volume adjustment.
- / – mute toggle.

Some gamepad hotkey combinations additionally provide reset, Quick Menu access, quitting the running title and quitting RetroArch back to the Playlists menu.

Save States

Save states are especially handy when revisiting arcade and early console games.

The K/L/N/P mappings provide direct keyboard control over:

- saving
- loading
- moving to the next state slot
- moving to the previous state slot

That avoids repeatedly opening RetroArch menus during play.

Special Help: Atari 2600 Controls

The Atari 2600 looks simple until an emulator has to reproduce all those physical switches from the original console.

The build maps those switches to modern controller buttons and provides on-screen notifications when they are changed.

The documented mappings include:

- SELECT – Atari Option/Select switch.
- START – Game Reset switch.
- L1 / R1 – set left/right Difficulty switches to A.
- L2 / R2 – set left/right Difficulty switches to B.
- Thumb Left / Thumb Right – switch TV type between color and black-and-white modes.

These settings can affect game modes, difficulty and even how individual cartridges behave.

Special Help: Atari 7800

The Atari 7800 also uses the difficulty switches in game-specific ways.

Depending upon the game, L1/R1 difficulty-switch settings can affect things such as:

- background music
- difficulty
- player count
- level selection

- landing gear
- time-outs
- weapon behavior
- game speed

If an Atari 7800 title appears to behave differently from memory, check those switch states before assuming something is wrong with the emulator.

Special Help: ColecoVision

The ColecoVision controller's numeric keypad makes emulation slightly unusual.

The configuration allows keyboard numbers 1 through 8 to start many games and maps the console keypad/dial functions across available RetroPad buttons.

Player 1 can also use keyboard 0 and 9 where appropriate, while Player 2 can make use of the numeric keypad.

If you launch a ColecoVision game and appear to be stuck on its selection screen, try the keyboard number keys before assuming the game hasn't started correctly.

Special Help: Commodore VIC-20, C64 and Amiga

This is where the build's attention to detail really earns its keep.

A Commodore computer is not a games console.

Its keyboard is part of the machine.

Because of this, Commodore emulation launches with Game Focus enabled, allowing the emulator core to receive keyboard and mouse events directly.

The side effect is that normal RetroArch keyboard hotkeys will not operate while Game Focus is enabled.

If you need RetroArch hotkeys:

Press Caps Lock to switch Game Focus OFF.

Press it again when you need the emulated machine to take control of the keyboard.

Changing Commodore Floppy Disks

Multi-disk games are supported with dedicated shortcuts:

- \ – eject or insert the currently selected floppy.
- [– select the previous floppy disk.
-] – select the next floppy disk.

This saves the considerable inconvenience of repeatedly entering RetroArch's disk-management menus.

Commodore Status Information

The Pi/logo key can toggle a bottom status bar displaying useful information such as:

- current joystick-port mapping
- floppy-drive activity
- screen refresh rate
- PAL 50Hz / NTSC 60Hz status

Joystick In The Wrong Port?

Commodore veterans will know exactly why this exists.

Some games expect the joystick to be connected to Port 1. Others expect Port 2.

The configuration provides a shortcut for swapping joystick Port 1 and Port 2 devices without diving through configuration menus.

On-Screen Commodore Keyboard

The controller's SELECT button can bring up an on-screen keyboard for Commodore systems.

That makes plenty of computer titles practical from the sofa without requiring a physical keyboard permanently attached to the Pi.

DOS Gaming Help

DOS games run through DOSBox-Pure.

The documented controller setup provides:

- Thumb Left – toggle the on-screen keyboard.
- Select – quit the running game.

An on-screen keyboard is particularly useful for titles which start with a text prompt, character-name entry or keyboard-based menu before becoming primarily joystick controlled.

RetroArch NetPlay

NetPlay is another major feature hiding inside this image.

RetroArch NetPlay attempts to recreate local multiplayer over the internet by continuously synchronizing multiple RetroArch instances running compatible content and emulator cores.

This isn't designed to emulate every original networking system.

It is primarily intended to take a game that originally expected several people around one machine and allow those controller inputs to come from remote RetroArch clients.

First Boot

The first boot can take longer than subsequent starts.

Let the machine finish its initial startup before assuming it has locked up.

Once it has settled, configure:

- network connectivity
- audio/sound
- Bluetooth devices
- display preferences where necessary

These can be managed through the desktop/system controls.

Default Login

The image is configured for automatic login in both desktop and console modes.

The supplied documentation gives the default account as:

User: pi

Password: Freedom

If the machine will be connected to an untrusted network or exposed beyond a private home environment, changing default credentials is sensible.

Switching Between Console And Desktop Environments

The system is deliberately more flexible than a traditional appliance image.

From the console menu, pressing the left Alt key can prompt for switching between running shell/terminal profiles.

The boot configuration and shell environment prepare the correct Raspberry Pi model-specific environment during startup.

Desktop mode provides broader Linux controls and, in the current release, is described as offering more comprehensive features and controls.

Why Some Settings Reset After Reboot

This is important, because otherwise users may think something is broken.

The RetroArch setup is deliberately designed so that many runtime changes are temporary.

When the system reboots, its base configuration can be restored.

This isn't an accident.

It is designed to stop experimental setting changes from gradually destroying a known-good gaming configuration.

The underlying setup uses a template configuration which is copied into the runtime RetroArch configuration, with additional playlist-specific configuration files applied afterwards to override appropriate values.

For people who simply want to play, this is excellent.

You can tweak something, make a mess, restart the machine and be much closer to the intended baseline again.

For Advanced Users: Editing The Base Configuration

Advanced Linux users aren't completely locked out.

The documentation suggests dropping to a shell and comparing the supplied configuration files:

```
ll *.cfg
diff default.cfg template.cfg | less
micro template.cfg
```

In other words, understand the project's configuration layering before permanently modifying it.

Automatic Raspberry Pi Model Detection

The startup scripts detect the Raspberry Pi model and prepare the environment accordingly.

This allows the build to make different performance and visual choices for Pi 4/400 and Pi 5/500-class hardware.

Past revisions have even used model detection to decide between faster and more accurate emulator-core choices where appropriate.

That is a much better approach than pretending every Pi behaves identically.

Performance Tweaks And Overclocking

The documentation includes example performance-oriented values for the supported Pi generations.

Overclocking is never guaranteed on every individual board, power supply and cooling arrangement, so anyone changing these values should understand the usual Raspberry Pi requirements around cooling, power quality and silicon variation.

Booting From NVMe On Raspberry Pi 5

The project also documents NVMe boot support.

The Raspberry Pi EEPROM configuration can be edited with:

```
sudo rpi-eeprom-config --edit
```

For a large build like this, a decent SSD or NVMe setup can make considerably more sense than relying on a low-end microSD card.

Updates: What The Upgrade Option Does

The console Playlists interface includes an upgrade option intended to apply ongoing enhancements to the current image baseline.

This has historically been used for updated:

- RetroArch runtimes
- configuration files
- scripts
- supporting files
- folder structures
- symbolic links

There is one extremely important distinction:

Newly added content is not necessarily included through the upgrade process.

The August 2026 documentation explicitly warns that added content may require re-imaging to obtain.

So if a future image release includes new games or large content changes, don't automatically assume running Upgrade will transform an older installation into a byte-for-byte equivalent of the newest image.

Using It As A Living-Room Retro Console

This is probably the easiest use case.

Connect the Raspberry Pi to a television, pair a controller, boot into the console or EmulationStation interface and treat the whole thing like an enormous multi-generation console.

The quick-picks and favorites features become especially useful here.

Using It In An Arcade Cabinet

The build is also particularly appealing for a Raspberry Pi arcade cabinet.

Useful cabinet-oriented features include:

- MAME and FinalBurn Neo support
- arcade playlists
- attract mode
- simple quick-pick lists
- controller abstraction

- arcade artwork and overlays
- crosshair support for suitable games
- easy volume control
- console-style shutdown and reboot
- automatic startup

A cabinet could boot straight into the gaming environment without exposing casual players to an ordinary Linux desktop.

Using It For Vintage Computer Exploration

This may actually be the project's strongest differentiator.

Plenty of retro images do consoles well.

Computers are harder.

The keyboard matters. The mouse matters. Joystick ports matter. Floppy disks matter. PAL and NTSC modes matter. Manuals matter.

This build has clearly spent significant development effort dealing with those realities.

For somebody interested in experiencing Commodore, Apple, Tandy, TI, DOS and other computer systems rather than merely browsing ROM names, this is a serious advantage.

Using It As A Retro Gaming Museum

With systems stretching from 1970s hardware through arcade machines, 8-bit micros, 16-bit consoles, 32-bit systems, Dreamcast, GameCube, PSP and more, the Pi can become a surprisingly capable interactive history of gaming.

Add manuals, artwork and EmulationStation presentation and it becomes something you can comfortably demonstrate to people who have never used the original hardware.

Using It For Multiplayer Nights

The project has several features that suit a games-room machine:

- party collections
- sports games
- four-player arcade games
- Dreamcast multiplayer
- Nintendo multiplayer titles
- local controller support
- NetPlay for remote multiplayer

A Raspberry Pi permanently connected to a large display could easily become the machine that gets switched on whenever friends come round.

Using It For Development And Experimentation

Unlike a locked-down console distribution, there is a real Debian Linux system underneath everything.

There is even a convenience script documented for fetching/building RetroArch and selected cores:

```
make-retroarch.sh
```

That makes the project interesting to developers and emulator enthusiasts as well as people who simply want to play.

Troubleshooting: Controller Not Detected

If a controller isn't responding:

- Return to the console menu if possible.
- Use the controller-list function to inspect connected USB and Bluetooth devices.
- Confirm Bluetooth is enabled if using a wireless controller.
- Check whether the operating system sees the device before assuming RetroArch is the problem.
- Reconnect or re-pair the controller.
- Restart the Pi after pairing if necessary.

The project has received repeated updates to its udev/input-device detection logic specifically for joysticks, keyboards and mice.

Troubleshooting: Keyboard Shortcuts Don't Work In Commodore Games

This is usually intentional.

Commodore games start with Game Focus enabled so the emulated computer receives keyboard input.

Press Caps Lock.

That switches Game Focus off and allows the RetroArch keyboard commands to work again.

Troubleshooting: Commodore Joystick Doesn't Control The Game

The game may expect the joystick in the other original Commodore joystick port.

Use the configured joystick-port swap command before modifying the emulator itself.

Troubleshooting: Multi-Disk Game Asks For Disk 2

For supported Commodore configurations:

- Use] to select the next disk.
- Use \ to eject/insert the selected disk.
- Use [if you need to go back to the previous disk.

Troubleshooting: ColecoVision Game Won't Start

Try keyboard numbers 1 through 8.

The original ColecoVision controller included a keypad and many games expect a number to be selected before gameplay begins.

Troubleshooting: A Setting Disappears After Restart

That may be deliberate.

Many runtime RetroArch settings are restored from the project's known-good configuration during startup.

If you want a permanent modification, investigate the template and appended configuration system instead of repeatedly changing the temporary runtime configuration.

Troubleshooting: First Boot Looks Slow

Give it time.

The documentation explicitly notes that first boot can take longer while the system settles.

Do not immediately pull the power simply because the initial start takes longer than later boots.

Troubleshooting: No Sound Or Wrong Output

Boot into the desktop environment and check the system audio controls.

Recent releases include WirePlumber mixer improvements intended to improve audio volume behavior during startup and switching from desktop mode.

With televisions and AV receivers, also confirm that the intended HDMI output is being selected.

Troubleshooting: Game Runs But Controls Are Strange

Before remapping everything globally, remember that RetroArch works through multiple layers:

Physical Controller → RetroPad → Emulator Core → Emulated Machine

A problem at one layer does not necessarily mean the entire controller profile is wrong.

For Atari, ColecoVision and vintage computers in particular, some apparently strange controls are actually reproductions of switches, keypads or ports on the original hardware.

Troubleshooting: NetPlay Cannot Connect

Check:

- host IP
- host NetPlay session status
- firewall
- UPnP
- TCP port 55435 forwarding
- Relay Server option
- matching content
- matching emulator core

For the best chance of a successful session, both sides should be running identical or suitably compatible game content and emulator cores.

Troubleshooting: EmulationStation Has Opened And I Want To Return

Use the configured kiosk quit control to exit EmulationStation and return to the console Playlists environment.

SELECT and START additionally expose more navigation options inside the frontend.

Frequently Asked Questions

► *Is this only RetroArch?*

No. RetroArch is central to the project, but the complete environment also includes Debian Linux, the custom console Playlists system, EmulationStation, standalone/supporting emulators where required, documentation integration and desktop functionality.

► *Does it work on Raspberry Pi 4?*

Yes. The current documentation lists Raspberry Pi 4 and 400-class systems as supported targets.

► *Does it work on Raspberry Pi 5?*

Yes, and the configuration contains model-specific Pi 5 optimization and NVMe-related settings.

► *What about Raspberry Pi 500?*

The Debian 13 generation has been tested/documented for Pi 500-class hardware, with additional Pi 500+ keyboard-lighting integration noted in the current changelog.

► *Can I use an NVMe SSD?*

Yes. Pi 5 NVMe and PCIe Gen 3 configuration is documented, along with EEPROM boot configuration.

► *Can I use a PS5 DualSense?*

Yes. DualSense detection and enhanced integration are specifically mentioned in the project documentation.

► *Does it support Bluetooth controllers?*

Yes. The console interface can list connected USB and Bluetooth controllers, and the underlying desktop provides Bluetooth controls.

► *Does it include EmulationStation?*

Yes. A customized EmulationStation configuration provides a graphical kiosk-style alternative to the project's console playlists.

► *Does it have game manuals?*

Manual links are available for supported games through the Bookshelf functionality and within suitable EmulationStation entries.

► *Does it support RetroAchievements?*

Yes. Compatible games can use a Cheevos/RetroAchievements account.

► *Can games be played online?*

Compatible local multiplayer games can use RetroArch NetPlay to synchronize players over a network or the internet.

► *Does NetPlay emulate original network or link-cable modes?*

No. The documented implementation is for recreating same-system local multiplayer through synchronized RetroArch instances rather than reproducing every original hardware networking technology.

► *Can it be used like a normal Linux PC?*

Yes. The build includes a Debian-based desktop environment as well as the console gaming interface.

► *Can I permanently modify RetroArch?*

Yes, but understand the configuration layering first. Runtime changes may intentionally reset on reboot.

► *Does the Upgrade option add every new game from future images?*

No. The current documentation specifically warns that added content is not necessarily part of the upgrade process. A newer image may need to be written again to obtain major content additions.

Who Is This Build For?

We can see several audiences absolutely loving this one.

► *The Raspberry Pi Owner With A Spare Pi*

If a Pi 4 or Pi 5 has been sitting in a drawer waiting for a purpose, this gives it a very, very big one.

► *The Retro Gamer*

Hundreds of curated favorites across dozens of machines without having to build every emulator configuration individually.

► *The Arcade Cabinet Builder*

The console interface, quick-picks, artwork, attract mode, MAME support and controller flexibility are all useful in a permanent cabinet.

► *The Vintage Computer Fan*

This is arguably where the image becomes genuinely special. Commodore, DOS, Apple, Tandy and other computer systems receive attention to keyboards, disks and manuals rather than being treated like consoles.

► *The Linux Tinkerer*

There is still a Debian installation underneath it all, complete with shells, configuration files and build scripts to investigate.

► *The Collector Who Wants One Box Under The Television*

Atari, Commodore, Nintendo, Sega, Sony, arcade, computers and handhelds all living behind one HDMI connection is difficult to argue with.

A Few Sensible Recommendations Before You Start

- Use a good-quality Raspberry Pi power supply
- Use decent cooling, particularly if running performance-oriented Pi 5 settings
- Use a quality storage device rather than the cheapest microSD card available
- Consider NVMe or SSD storage on Pi 5 for a collection of this size
- Keep the HELP documentation handy
- Pair your controller before assuming a game mapping is broken
- Learn the Caps Lock Game Focus shortcut if you play computer titles
- Remember the Commodore disk-change shortcuts
- Remember that Atari switches are part of the original machine behavior
- Don't assume a temporary RetroArch setting will survive reboot
- Back up anything you personally modify

Final Thoughts

There are Raspberry Pi retro images that concentrate on looking pretty.

There are others that concentrate on packing in enormous libraries.

And there are enthusiast builds that are incredibly configurable but expect the user to spend their first weekend fixing controllers and editing configuration files.

This project is interesting because it attempts to sit somewhere in the middle of all three.

It has the huge multi-system appeal.

It has the graphical EmulationStation experience.

It has an immediately usable console menu.

It has genuinely useful touches such as RetroAchievements, manuals, save states, attract mode, Bluetooth support and NetPlay.

But it also retains the wonderfully nerdy side of retro computing: Commodore floppy management, joystick-port switching, Atari hardware switches, emulator cores, model-specific Raspberry Pi configuration, NVMe support, Linux shells and configuration layering.

That combination is what makes it considerably more than another Raspberry Pi RetroArch image.

With nearly 128GB of emulation and system content, around 50GB of additional multimedia, support spanning multiple decades of gaming hardware and years of ongoing configuration work behind it, there is an enormous amount here to explore.

Console gamers can simply pick up a controller and play.

Vintage-computer fans can dig into machines they grew up with.

Arcade builders can turn it into the center of a cabinet.

Linux enthusiasts can open a terminal and start pulling it apart.

And everyone else can lose an unreasonable number of hours jumping from Atari to Amiga, from MAME to Mega Drive, from PlayStation to Dreamcast, and then somehow end the evening playing Bubble Bobble again.

Exactly as a great retro build should.